Revolution Crossroads — what kind of world is this?

Exploratory notebook (companion to `notes/DIRECTION.md`)

Built on the class notebook `PGDV_5200_text_processing.ipynb`: streaming `Dataset` → pandas → sklearn / BERTopic. No SQL.

Datasets (all `RevolutionCrossroads/…` on Hugging Face):

- `RevX_OCR_results` — the AI OCR run. `page_level` (one row = one page) and `block_level` (one row = one paragraph/header/table). No date/title/place columns; for newspapers they're packed into `filename` = `{lccn}_{YYYY-MM-DD}_{edition}.pdf`. `page_number` is 0-based.
- `RevX_NER_Entity_Occurrences` — 24M entity mentions (person, place, organization, date…) extracted by an LLM, each with its surrounding `block_text`. Configs `chronam`, `pension`, `si`, `all`. No relations: we build those.

Sections: 0 setup · 1 pages · 2 regimes vocabulary · 3 search · 4 blocks (two registers) · 5 clustering · 6 zeal test · 7 entities & relations · 8 export

0. Setup

```
import re, os, itertools, collections
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from datasets import load_dataset

from sklearn.decomposition import LatentDirichletAllocation
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer

pd.set_option("display.max_colwidth", 200)
os.makedirs("data", exist_ok=True)

# Hugging Face token: Colab → key icon → add secret HF_KEY (or whatever name you used)
try:
    from google.colab import userdata
    HF_KEY = userdata.get("HF_KEY")
except Exception:
    HF_KEY = os.environ.get("HF_TOKEN")   # local: `huggingface-cli login` sets this

OCR = "RevolutionCrossroads/RevX_OCR_results"
NER = "RevolutionCrossroads/RevX_NER_Entity_Occurrences"
LOC = "RevolutionCrossroads/loc_chronicling_america_1770-1810"

FILENAME_RX = r"^(?P<lccn>sn\d+)_(?P<issue_date>\d{4}-\d{2}-\d{2})_(?P<edition>\d+)\.pdf$"

def stream(name, config, n, filt=None, seed=1010, buffer=5000, columns=None):
    """Stream n rows of a HF dataset into a DataFrame, in the class idiom."""
    ds = load_dataset(name, name=config, split="train", streaming=True, token=HF_KEY)
    if columns:
        ds = ds.select_columns(columns)
    if filt:
        ds = ds.filter(filt)
    ds = ds.shuffle(seed=seed, buffer_size=buffer).take(n)
    return pd.DataFrame(list(ds))

def add_newspaper_metadata(df):
    """Parse lccn / issue_date / edition / year from newspaper filenames (NA for pensions)."""
    parts = df["filename"].str.extract(FILENAME_RX)
    df["lccn"] = parts["lccn"]
    df["issue_date"] = pd.to_datetime(parts["issue_date"], errors="coerce")
    df["year"] = df["issue_date"].dt.year.astype("Int64")
    if "page_number" in df:
        df["page"] = df["page_number"] + 1
    return df
```

1. Pages: a first look at the newspapers

`page_level`, newspapers only. `N_PAGES` small to start; raise it when the pipeline works.

```
N_PAGES = 70000

pages = stream(OCR, "page_level", N_PAGES, filt=lambda e: e["collection"] == "chronicling_america")
pages = add_newspaper_metadata(pages)
pages["text"] = pages["new_ocr_text"].fillna("")
pages = pages[["lccn", "issue_date", "year", "page", "text"]]

print(len(pages), "pages from", pages["lccn"].nunique(), "newspapers,",
      pages["issue_date"].min().date(), "to", pages["issue_date"].max().date())
pages.head(3)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fa
Resolving data files: 100%                                  27/27 [00:00<00:00, 528.38it/s]
70000 pages from 110 newspapers, 1770-01-05 to 1810-12-31
```

	lccn	issue_date	year	page	text
0	sn84024736	1807-01-10	1807	4	FROM THE PALLADIUM EXTRA. LEGISLATURE OF KENTUCKY. IN THE HOUSE OF REPRESENTATIVES. TUESDAY, DECEMBER 20, 1806. THE DEPOSITION OF HARRY INNES.\n[CONCLUDED]\nThis deponent since making this communi...
1	sn83030137	1787-09-18	1787	2	Extract from an Essay on the Means of Promoting Federal Sentiments in the United States, by a Federal Spectator, published in the Independent Gazetteer, at Philadelphia.\nA well ordered militia is...
2	sn84024013	1809-02-20	1809	3	valuable cargo of wine, oil, flour, beef, dry goods,\nSince writing the above, our minds have been still further relieved from gloomy apprehensions. H. M. ship Circe, captain Collier, arrived last...

Next steps: Generate code with pages | New interactive sheet

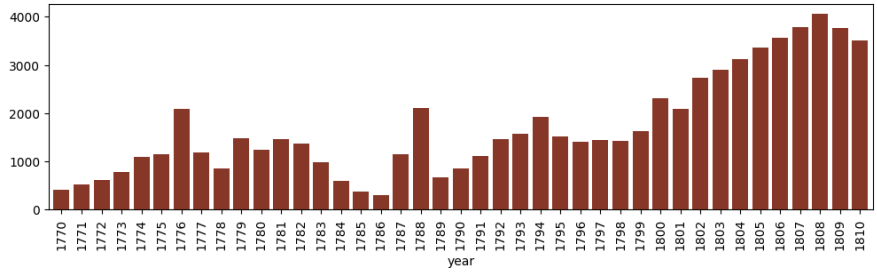
```
# Titles and places live in the LOC dataset; borrow a lookup (one row per lccn)
papers = (stream(LOC, "default", 20000, columns=["lccn", "newspaper_title", "place_of_publication"])
          .drop_duplicates("lccn").rename(columns={"place_of_publication": "place"}))
pages = pages.merge(papers, on="lccn", how="left")
pages.groupby(["newspaper_title", "place"]).size().sort_values(ascending=False).head(10)
```

README.md: 100% 23.7k/23.7k [00:00<00:00, 2.05MB/s]

newspaper_title	place	0
The national intelligencer and Washington advertiser.	Washington City [D.C.]	4989
The New Hampshire gazette	Portsmouth [N.H.]	3306
Gazette of the United States, & Philadelphia daily advertiser.	Philadelphia [Pa.]	3211
The New-Hampshire gazette and general advertiser	Portsmouth [N.H.]	1669
Gazette of the United-States.	New-York [N.Y.]	1643
The New-Jersey gazette	Burlington [N.J.]	1520
The daily advertiser	New York [N.Y.]	1337
The New-York gazette, and the weekly Mercury	New-York [N.Y.]	1100
The New-Hampshire gazette and historical chronicle	Portsmouth, N.H.	1034
National gazette.	Philadelphia [Pa.]	746

dtype: int64

```
# The archive is uneven: every count later is divided by this
per_year = pages.groupby("year").size().rename("pages")
per_year.plot.bar(figsize=(12, 3), color="#8a3b2a", width=0.8); plt.show()
```



2. The world's three native regimes

Sensibility, paranoia, enthusiasm: the period's own names for sanctioned, feared and surplus feeling (see DIRECTION.md). `CountVectorizer(vocabulary=…)` counts only our words; `ngram_range=(1,3)` allows phrases. Edit freely — choosing these is the argument.

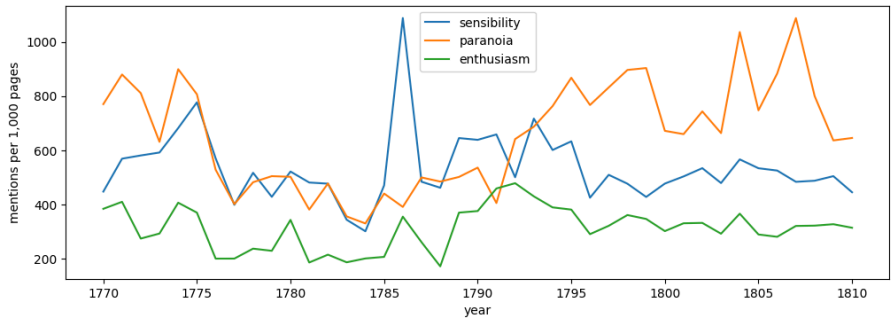
```
REGIMES = {
    "sensibility": ["sympathy", "sympathetic", "tender", "affecting", "sensibility", "feeling heart",
                    "tears", "compassion", "benevolence", "humanity"],
    "paranoia":    ["conspiracy", "design", "designs", "plot", "faction", "incendiary", "demagogue",
                    "intrigue", "secret", "jesuitical"],
    "enthusiasm":  ["enthusiasm", "enthusiast", "zeal", "zealot", "fanatic", "fanaticism", "frenzy",
                    "phrenzy", "mob", "tumult", "rage"],
}
VOCAB = [w for ws in REGIMES.values() for w in ws]

vec = CountVectorizer(vocabulary=VOCAB, ngram_range=(1, 3), lowercase=True)
counts = pd.DataFrame(vec.fit_transform(pages["text"]).todense(), columns=vec.get_feature_names_out())
counts.sum().sort_values(ascending=False).head(15)
```

	0
design	12369
secret	12173
humanity	10922
zeal	10054
tender	8356
designs	7092
faction	6542
conspiracy	4904
rage	3758
tears	3701
benevolence	3305
sensibility	3199
affecting	2998
plot	2902
mob	2763

dtype: int64

```
# Mentions per 1,000 pages, per regime, per year
regime_counts = pd.DataFrame({r: counts[ws].sum(axis=1) for r, ws in REGIMES.items()})
regime_counts["year"] = pages["year"].values
rate = regime_counts.groupby("year").sum().div(per_year, axis=0).mul(1000).round(1)
rate.plot(figsize=(12, 4)).set_ylabel("mentions per 1,000 pages"); plt.show()
rate.tail(10)
```



year	sensibility	paranoia	enthusiasm
1801	503.4	659.5	330.5
1802	533.9	743.3	331.9
1803	479.0	663.3	292.0
1804	566.3	1036.3	365.9
1805	533.6	747.3	289.2
1806	525.1	883.2	280.6
1807	483.5	1087.6	320.7
1808	487.5	800.0	321.8
1809	504.4	636.3	327.1
1810	445.0	645.3	314.0

Start coding or generate with AI.

3. Read before you count

`search()` shows every regex match with the words around it. Signal vs noise ("leviathan" the book vs the ship) is decided here, by reading.

```
def search(pattern, context=60, limit=30, df=None, col="text"):
    df = pages if df is None else df
    rx = re.compile(rf"(?is).{{0,{context}}}(?:{pattern}).{{0,{context}}}")
    rows = []
    for _, r in df[df[col].str.contains(pattern, case=False, regex=True, na=False)].iterrows():
        for m in rx.findall(r[col]):
            rows.append({**{k: r[k] for k in ("issue_date", "lccn", "collection", "register") if k in r},
                         "snippet": re.sub(r"\s+", " ", m)})
            if len(rows) >= limit:
                return pd.DataFrame(rows)
    return pd.DataFrame(rows)

search(r"\bzeal")
```

	issue_date	lccn	snippet
0	1797-11-21	sn83025525	ts who are skilled in kindling up every where the murderous zeal of homicidal pity.—The National Tribune no longer resounded
1	1809-12-23	sn84024013	ing his high character, his great talents, and unquestioned zeal for the honor and prosperity of his country; he could not b
2	1798-07-10	sn83025525	ty. In this all important conflict, we expect the ready and zealous cooperation of the free and enlightened people of Americ
3	1798-07-10	sn83025525	on to expect, in this all important conflict, the ready and zealous cooperation of the free and enlightened people of Americ
4	1800-09-26	sn84026272	small confidence in my slender abilities, but animated with zeal for the improvement of those who may wish to learn French f
5	1800-09-16	sn84026272	small confidence in my slender abilities, but animated with zeal for the improvement of those who may wish to learn French f
6	1806-08-13	sn85025815	At least this appears to be the case from the activity and zeal displayed by his placemen in that commonwealth. The adminis
7	1779-12-25	sn83021127	Majesty have an opportunity given them, to signalize their zeal for your service, and for the benefit of their country, the
8	1792-02-15	sn83025587	ces will direct our mutual efforts, and we depend that your zeal for the public good will equal ours. “We were ignorant at o
9	1801-10-14	sn83045242	ted it with a great deal of patience, and have manifested a zeal which is above all eulogy, to the honor of his majesty's ar
10	1801-10-14	sn83045242	great obligations to quarter master Gen. Autruther, for the zeal and ability he displayed for the public service under the m
11	1801-07-28	sn84023378	the passions, he is never transported into excesses by the zeal of his friends, or the virulence of his enemies. His object
12	1801-07-28	sn84023378	f Aboukir. I have many obligations to Col. Spencer, for the zeal, activity, and military talents, which he has displayed in
13	1801-07-28	sn84023378	order of things in Germany. The former has lost much of her zeal and friendship for the French Republic, and the latter has
14	1801-12-16	sn83045242	heck. To captain Cochrane I feel extremely obliged, for his zeal and constant readiness to afford every assistance in his po
15	1801-12-16	sn83045242	r side, and calls upon me to pay before you the unremitting zeal and attention of lieut. col. Darby, who, with the first bat
16	1800-03-08	sn83025544	justly due to seamen, & soldiers who have manifested their zeal, activity and bravery so conspicuously, in support of the h
17	1797-08-31	sn83025881	carry it on, with every care, attention and dispatch that a zeal to oblige the public can possibly exert. The above company,
18	1787-02-10	sn83025586	is owing, in a measure, to the patience and obedience, the zeal and the fortitude discovered in our troops, to whom too muc
19	1787-07-26	sn83030565	racious speech, expressive of his entire approbation of the zeal of his lords and commons, in the business he had recommende
20	1772-09-24	sn84024739	he most whimsical System imaginable. He defends, with great Zeal, the Opinion of Doctor Smith, that our Species has not, sin
21	1781-07-02	sn83021127	s, and I hope the success of my efforts will be equal to my zeal, and the important objects of the trust you have reposed in
22	1794-11-17	sn84026271	y of the Empire, which has been contrived with all possible zeal, has not been effected at this moment when two thirds of th
23	1794-11-17	sn84026271	brave fellow citizens. We need not excite or support their zeal; their love of their country wants no encouragement; but it
24	1794-11-17	sn84026271	afflicting sorrow that we mention the sudden death of that zealous patriot and truly respectable citizen, Mr. Robert Elliot
25	1808-08-19	sn84024736	ority, that Gen. Moore's expedition was destined to re-take Zealand and Copenhagen, and give it to the king of Sweden; & tha
26	1808-08-19	sn84024736	to 3000 at least, had contrived to pass the Bells and reach Zealand. A camp consisting of from 8 to 10,000 Danish troops, wa
27	1800-09-16	sn84026272	the officers and Administrators who shall have served with zeal, and will point out to public opinion those who have shown
28	1787-10-30	sn83030137	, and by the bitterness of their invectives. An enlightened zeal for the energy and efficiency of government will be stigmat
29	1787-10-30	sn83030137	erous ambition more often lurks behind the specious mask of zeal for the rights of the people, than under the forbidding app

```
search(r"\bdesign")
```

	issue_date	lccn	snippet
0	1807-01-10	sn84024736	holic majesty's provinces of East and West Florida shall be designated by a line commencing on the Mississippi at the mouth of
1	1807-01-10	sn84024736	boundary of his N o s provinces of E. & W. Florida shall be designated by a line beginning at the same point on the Mississipp
2	1787-09-18	sn83030137	a kind of key to Utrecht, where the Prince certainly has a design of commencing hostilities. The Patriots (we use their own p
3	1798-05-23	sn82014782	are verbally informed that the French still shew a serious design of invading England, as gun boats to the number of about 10
4	1780-12-20	sn83025471	s, lest they may be used as a pretext to cover some hostile design. Much is to be apprehended from the profound policy of the
5	1780-12-20	sn83025471	have joined this league, which, though it should be really designed, as it is pretended, for the protection of commerce, is y
6	1810-11-27	sn83025588	for supporting their extensive establishments, in the pious design of keeping down, by under-bidding, domestic manufacturers.
7	1810-11-27	sn83025588	efore, to prevent them from realizing their vile and unjust designs.—With equal care and vigilance, keep aloof from all preten
8	1801-03-20	sn83045242	n Branch. They are spacious and convenient, one of which is designed for stage passengers and travellers, the other for the ac
9	1785-01-21	sn83025586	nce, it concerns the PUBLICK to inquire into its nature and design;—that if it be innocent it may have the common protection;
10	1783-05-13	sn83021127	ance to New-York safe, speedy and commodious, it is hoped a design so well adapted to public convenience, will meet with publi
11	1807-03-13	sn84024736	rrect, so far as it respects his military character and his designs upon foreign nations. But it does him no sufficient justic
12	1807-03-13	sn84024736	b. 17. SIR, The enclosed resolution passed yesterday. It is designed to ascertain if the number of seamen allowed by law for t
13	1803-07-29	sn83045242	nstrances only, and orders to give up our conquests. 3. The designs of France growing ripe for seizing Egypt, Government was d
14	1797-01-30	sn83025881	to conclude with A GRAND PAGEANT. The scene of the Volcano designed by J. Richards, Esq. R. A. principal Scene Painter to Cov
15	1797-01-30	sn83025881	ent-Garden Theatre. The rest of the Scenery designed by Mr. Milbourne, and the whole executed by him and under
16	1797-01-30	sn83025881	his direction. Machinist, Mr. Lenthall. The dresses designed by Mr. Gibbons. To which will be added, the musical drama
17	1788-09-06	sn82014784	ected places or apprehend suspected persons without special designation of the objects of search, arrest or seizure, should no
18	1796-10-08	sn82014782	a considerable degree of esteem; but these being originally designed for young men who have been initiated in classical studie
19	1805-10-11	sn84024012	they hold north of the Tennessee river, except a small slip designed to include S. W. Point. It is supposed the conference wil
20	1801-05-15	sn83045242	elates to aerial machinery, but differing throughout in its design and conduct, this fascinating volume commences. Wildness of
21	1775-07-03	sn83021126	edious essay, whereas I ought to confine myself to the main design and purpose of this letter, which is to guard you and your
22	1775-07-03	sn83021126	you as he has deceived himself; I do not say he will do it designedly. I do not think him capable; but his mind is so totally
23	1809-01-17	sn84024013	continuance of the embargo, to such foreign ports as may be designated by law, for the sole purpose of bringing in cargoes of
24	1799-11-13	sn83021629	of the fall of man, whether literally true, or an allegory designed to represent the beginning and progress of evil in the hu
25	1799-11-13	sn83021629	ned how much the enemy had suffered, he renounced his first design, and determined to cover the Genoese territory. We shall re
26	1799-11-13	sn83021629	marched with the main body towards Asti, which showed their design to besiege Coni, the only strong place of which we are in p
27	1804-08-23	sn83021188	omplishment of their purpose, con- victs them of iniquitous design. Let it be re- membered that the leading postulate of their
28	1804-08-23	sn83021188	1st, that those to whom it is offered with the inten- tion (design in this case would be a more appro- priate word) of influen
29	1806-10-18	sn84024710	. A Table of Fines, Forfeitures, Penalties and Amercements, designating such as are to the use of the Commonwealth; such as go

4. Blocks: the two transmissions

`block_level`, prose blocks only (`block_type == "Text"`, 200–2,000 chars). Sampled **separately per register** — pensions outnumber newspapers ~30:1, so a mixed sample is 95% pension.

Pension files have no date in the filename; `year` is NA for them until we join the NARA metadata (to do).

```
N_PER_REGISTER = 3000
REGISTER = {"chronicling_america": "town", "rev_war_pension": "state"}

def is_prose(b):
    t = b["block_text"]
    return b["block_type"] == "Text" and t is not None and 200 <= len(t) <= 2000

frames = []
for coll, reg in REGISTER.items():
    df = stream(OCR, "block_level", N_PER_REGISTER,
                filt=lambda b, c=coll: b["collection"] == c and is_prose(b),
                columns=["filename", "collection", "page_number", "block_id", "block_type", "block_text"])
    df["register"] = reg
    frames.append(df)

blocks = pd.concat(frames, ignore_index=True)
blocks = add_newspaper_metadata(blocks)
blocks["text"] = blocks["block_text"]
print(blocks["register"].value_counts())
print("newspaper issues:", blocks.loc[blocks.register == "town", "filename"].nunique(),
      "| years:", blocks["year"].min(), "–", blocks["year"].max())
```

```
Resolving data files: 100%                                                  27/27 [00:00<00:00, 2108.63it/s]
Resolving data files: 100%                                                  27/27 [00:00<00:00, 2116.08it/s]
Resolving data files: 100%                                                  27/27 [00:00<00:00, 932.39it/s]
Resolving data files: 100%                                                  27/27 [00:00<00:00, 1265.65it/s]
register
town     3000
state    3000
Name: count, dtype: int64
newspaper issues: 131 | years: 1770 – 1794
```

```
# Read a few from each register before doing anything else
for reg in ["town", "state"]:
    print(f"=== {reg} ===")
    for t in blocks[blocks.register == reg]["text"].sample(3, random_state=1):
        print(re.sub(r"\s+", " ", t)[:400], "\n")
```

```
=== town ===
FOR SALE, A Valuable Tract of LAND, in this county, containing about eight hundred acres, one fourth of which is low grounds, of excellent quality; the other

THIS little book is written in a style altogether novel, and is adapted to all capacities, as well as to all classes of people, merchants, mechanics and farme

On the 21st November, Wurmser was attacked by 25,000 French. Both sides fought with fury from morning till four in the afternoon. The field of battle was cove

=== state ===
To the 7th He says that the names of the persons to whom he is known in his present neighbourhood and as he believes can testify as to his character for verac

I Elihu Warner of Killingly in the County of Windham and State of Connecticut aged 76 years & over & being duly sworn according to law do testify and say that

I Gardner Preston of South Hadley in the County of Hampshire & Commonwealth of Massachusetts, aged 83 years, testify & say – that in July 1780. I enlisted at
```

5. Clustering: what sounds alike (optional, slow on CPU)

BERTopic groups passages by semantic similarity. The model decides what *sounds* alike; you decide what the likeness *means*, by reading ten per cluster and naming the rule.

```
!pip install -q bertopic sentence-transformers
```

```
from bertopic import BERTopic
from sentence_transformers import SentenceTransformer

embedder = SentenceTransformer("all-MiniLM-L6-v2")
topic_model = BERTopic(embedding_model=embedder, min_topic_size=20, verbose=True)
blocks["cluster_id"], _ = topic_model.fit_transform(blocks["text"].tolist())
topic_model.get_topic_info().head(15)

# read ten from one cluster, then name it in your own words
for t in blocks[blocks.cluster_id == 0]["text"].head(10): print(t[:300], "\n---")
```

```
modules.json: 100%                                                  349/349 [00:00<00:00, 29.0kB/s]
config_sentence_transformers.json: 100%                             116/116 [00:00<00:00, 11.7kB/s]
README.md: 100%                                                     10.5k/10.5k [00:00<00:00, 942kB/s]
sentence_bert_config.json: 100%                                     53.0/53.0 [00:00<00:00, 4.39kB/s]
config.json: 100%                                                   612/612 [00:00<00:00, 50.9kB/s]
model.safetensors: reconstructing file: 100%                        90.9MB / 90.9MB, 8.84MB/s
model.safetensors: downloading bytes:                               85.0MB, 8.26MB/s
Loading weights: 100%                                               103/103 [00:00<00:00, 3439.38it/s]
tokenizer_config.json: 100%                                         350/350 [00:00<00:00, 32.7kB/s]
vocab.txt: 100%                                                     232k/232k [00:00<00:00, 7.26MB/s]
tokenizer.json: 100%                                                466k/466k [00:00<00:00, 18.8MB/s]
special_tokens_map.json: 100%                                       112/112 [00:00<00:00, 8.31kB/s]
config.json: 100%                                                   190/190 [00:00<00:00, 17.4kB/s]
2026-10-27 10:11:55,924 - BERTopic - Embedding - Transforming documents to embeddings.
Batches: 100%                                                       188/188 [00:09<00:00, 45.78it/s]
2026-10-27 10:12:05,198 - BERTopic - Embedding - Completed ✓
2026-10-27 10:12:05,199 - BERTopic - Dimensionality - Fitting the dimensionality reduction algorithm
2026-10-27 10:12:35,090 - BERTopic - Dimensionality - Completed ✓
2026-10-27 10:12:35,091 - BERTopic - Cluster - Start clustering the reduced embeddings
2026-10-27 10:12:35,349 - BERTopic - Cluster - Completed ✓
2026-10-27 10:12:35,356 - BERTopic - Representation - Fine-tuning topics using representation models.
2026-10-27 10:12:35,838 - BERTopic - Representation - Completed ✓
Genoune and Guadet had the fervility to publish, at different times, that Louis XVI. had commanded the Swiss not to fire upon the people. From that time, the
---
Their passions have mounted to a height unknown in former wars. In such a state, it is but a word and a blow. In the bosom of America we have a faction, the
---
In pursuing the various and weighty business of the present session, I indulge the firmest persuasion, that your consultations will be equally marked with wi
---
As A. B. can scarcely believe himself to be "a prodigy of genius," notwithstanding the commendation of the eminent Gracchus, he is still in the dark about tha
---
From the whole then of the debate, without going into comparison of one nation and her conduct towards us, with another, which sometimes led into invidious d
---
In committee of the whole on Mr. Madison's resolutions. The motion made yesterday by Mr. Fitzsimons, for striking out the words, "Nations not having a commer
---
That while the paper credit of those colonies subsisted and was a legal tender, no instance can be given where bills of exchange and sterling debts were ever
---
Mr. Smilie said that the gentleman last speaking, had been uniformly opposed to opening the doors. The resolutions now proposed to be made public, are agreeal
---
was first agitated, at New-York, that such regulations then would not be well timed, considering that the execution of the treaty with Great Britain was yet
---
The French attacked us in all points at once, with from 70,000 to 80,000 men, commanded by General Jourdan; Gen. Dupont commanded their right.—Levassour and
---
```

```
if "cluster_id" not in blocks:
    blocks["cluster_id"] = -1   # placeholder until the cell above has run
```

6. The zeal test

Hypothesis: enthusiasm is an **accusation** in the newspapers and a **credential** in the pension files. Self-attributed vs other-attributed, per 1,000 passages, by register. `the zeal` is ambiguous: read those.

```
ZEAL = {
    "self":  r"\b(?:my|our)\s+(?:zeal|ardou?r|enthusiasm|patriotism)\b",
    "other": r"\b(?:his|her|their)\s+(?:zeal|ardou?r|enthusiasm|fanaticism|ph?renzy)\b",
    "the":   r"\bthe\s+(?:zeal|ardou?r|enthusiasm)\b",
}
for k, rx in ZEAL.items():
    blocks[f"zeal_{k}"] = blocks["text"].str.contains(rx, case=False, regex=True)

zeal = blocks.groupby("register")[[f"zeal_{k}" for k in ZEAL]].mean().mul(1000).round(1)
zeal.columns = [f"{k}_per_1k" for k in ZEAL]
zeal
```

register	self_per_1k	other_per_1k	the_per_1k
state	0.0	0.0	0.0
town	0.7	0.7	3.3

Next steps: Generate code with zeal | New interactive sheet

```
search(ZEAL["the"], context=80, df=blocks)
```

	issue_date	lccn	collection	register	snippet
0	1794-02-03	sn83025878	chronicling_america	town	ave translated my memorial into English, in order to lose no time, and to serve the zeal of a body composed of members, none of whom are ignorant of the value of the Fr
1	1794-02-22	sn83025878	chronicling_america	town	a treasury, the raising of 800,000 youth for the defence of the frontiers, and the ardour of all Frenchmen in the common cause, ought to convince the other powers of Eur
2	1794-02-13	sn83025878	chronicling_america	town	y the gentleman from Virginia. When he looked at that gentleman and attended to the zeal he dis- covered, in what he thought so essential to the welfare of his country,
3	1794-01-10	sn83025878	chronicling_america	town	flight, but they are pursued on all sides, and cannot possibly escape, not only the ardor of our troops, but also that of the people of Lyons, enraged at their having be
4	1794-03-22	sn83025878	chronicling_america	town	he assault. The rain and the most dreadful weather could not for a moment abate the ardor of the Republicans. Let not the services which the representatives of the peopl
5	1792-04-21	sn82014782	chronicling_america	town	e interior Indians and those towards the sea-coast parts. So great however, was the ardour of the enterprising Mr. Stuart to attain his object (the exploring the continen
6	1794-01-04	sn83025878	chronicling_america	town	mies who rise up against them in every shape and in all parts of the world, but the zeal of their agents at home & abroad shall be as constant as the invincible courage
7	1794-03-22	sn83025878	chronicling_america	town	mpletely. Never was a service performed with more regularity and exactness; but the ardour and impetuosity of the troops (instead of forming on the height where the batte
8	1794-01-11	sn83025878	chronicling_america	town	employ all the means which he derives from the resources of his dominions, from the zeal and affection of his people, and from the unquestionable justice of the cause.
9	1793-01-19	sn83025887	chronicling_america	town	us, that he is "waging war against the whole hell of monarchy," we must excuse the ardor of his attack, and pardon him for showing so little coolness during the heat of

7. Entities and relations

The NER dataset has entities but no relations. We build three kinds, cheapest first:

1. **The passageway** — persons named in *both* registers. Someone who appears in a 1777 newspaper and a 1832 pension declaration is the project's storyteller.
2. **Co-occurrence** — two entities named in the same block. Weak ("mentioned together") but it is the path of people.
3. **Typed relations by pattern** — `widow of`, `served under`, `late of`: the pension formulae are written by law, so regex gets high precision.

Each NER row carries its `block_text`, so no join back to the OCR is needed. `N_NER` rows per config is a sample; a full pass over 24M rows takes a while in Colab — do it once you know the queries work.

```
N_NER = 200_000          # per config; raise for real numbers
NER_COLS = ["collection", "record_id", "page", "window_id", "block_id", "entity_text",
            "entity_text_normalised", "entity_type", "confidence", "block_text"]

def ner_sample(config, n=N_NER, types=("person", "geographic_location", "organization"), min_conf=0.8):
    return stream(NER, config, n, columns=NER_COLS,
                  filt=lambda e: e["entity_type"] in types and e["confidence"] >= min_conf)

ner_news = ner_sample("chronam")
ner_pens = ner_sample("pension")
print(len(ner_news), "newspaper mentions;", len(ner_pens), "pension mentions")
ner_news["entity_type"].value_counts()
```

```
README.md: 100%                                          43.5k/43.5k [00:00<00:00, 2.70MB/s]
200000 newspaper mentions; 200000 pension mentions
```

entity_type	count
person	95508
geographic_location	85795
organization	18697

dtype: int64

7.1 The passageway: persons in both registers

```
def person_counts(df):
    p = df[df.entity_type == "person"]
    return p.groupby("entity_text_normalised").agg(mentions=("record_id", "size"),
                                                   documents=("record_id", "nunique"))

pn, pp = person_counts(ner_news), person_counts(ner_pens)
passage = pn.join(pp, lsuffix="_news", rsuffix="_pension", how="inner")

# drop single surnames and the John Smiths: need 2+ tokens, not absurdly common
passage = passage[passage.index.str.count(" ") >= 1]
passage = passage[(passage.mentions_news.between(2, 200)) & (passage.mentions_pension.between(1, 200))]
passage = passage.sort_values(["documents_pension", "mentions_news"], ascending=False)
print(len(passage), "candidate people in both registers")
passage.head(25)
```

```
888 candidate people in both registers
```

entity_text_normalised	mentions_news	documents_news	mentions_pension	documents_pension
a. b.	100	26	128	101
c. d.	40	24	63	63
j. edwards	20	19	34	28
general washington	18	15	33	28
john davis	47	44	50	24
john smith	150	132	49	21
mr. edwards	10	10	23	18
lord cornwallis	4	4	23	18
george washington	116	91	28	17
john turner	7	7	139	17
george taylor	8	8	16	15
john bailey	11	11	41	14
captain smith	4	4	15	13
general gates	2	2	16	13
john tyler	4	4	197	12
john hancock	6	5	14	11
john brown	55	46	18	10
william smith	57	47	13	9
john warner	27	24	19	9
gen. washington	22	21	12	9
thomas turner	2	2	31	9
john moore	69	66	18	8
john clark	34	34	23	8
william lloyd	11	10	8	8
david smith	8	8	11	8

Next steps: Generate code with passage | New interactive sheet

```
# Read the contexts for one candidate before believing the name match
def contexts(name, df, n=5):
    rows = df[(df.entity_text_normalised == name)].drop_duplicates("window_id").head(n)
    for _, r in rows.iterrows():
```

```
        print(f"[{r.collection} {r.record_id} p{r.page}]", re.sub(r"\s+", " ", r.block_text)[:400], "\n")

if len(passage):
    name = passage.index[0]
    print("=== newspapers ==="); contexts(name, ner_news)
    print("=== pensions ===");   contexts(name, ner_pens)
```

```
=== newspapers ===
[chronam sn82014782_1791-11-26_2 p4] "I, A. B. of \_\_\_\_\_ county of \_\_\_\_\_ state of \_\_\_\_\_ do hereby constitute and appoint C. D. of \_\_\_\_\_ my

[chronam sn82014782_1791-11-05_2 p4] "I, A. B. of county of state of do hereby constitute and appoint C. D. of my lawful attorney, to receive in my behalf of

[chronam sn82014782_1791-12-17_2 p4] "I, A. B. of — county of — state of — do hereby constitute and appoint C. D. of — my lawful attorney, to receive in my be

[chronam sn82014782_1791-11-19_2 p4] "I, A. B. of county of State of do hereby constitute and appoint C. D. of my lawful attorney, to receive in my behalf of

[chronam sn82014782_1791-11-12_2 p4] "I, A. B. of county of state of do hereby constitute and appoint C. D. of my lawful attorney, to receive in my behalf of

=== pensions ===
[pension 111404146 p29] “Personally appeared before me, the undersigned, a Justice of the Peace, &c. A. B. who, being duly sworn, deposeth and saith, that, by

[pension 111404274 p12] (k) We, A. B., a Clergyman, residing in the and C. D. residing in (the same) hereby certify, that we are well acquainted with , who ha

[pension 111404274 p13] "Personally appeared before me, the undersigned, a Justice of the Peace, &c. A. B. who, being duly sworn, deposeth and saith, that, by

[pension 111404239 p27] "Personally appeared before me, the undersigned, a Justice of the Peace, &c. A. B. who, being duly sworn, deposeth and saith, that, by

[pension 111405599 p6] “Personally appeared before me, the undersigned, a Justice of the Peace, &c. A. B. who, being duly sworn, deposeth and saith, that, by
```

7.2 Co-occurrence: who is named beside whom

```
def cooccurrence(df, group="window_id", min_weight=2, types=("person", "geographic_location", "organization")):
    """Edges between entities that share a text window. Returns source, target, weight, types."""
    d = (df[df.entity_type.isin(types)]
           .drop_duplicates([group, "entity_text_normalised"])
           [[group, "entity_text_normalised", "entity_type"]])
    edges = collections.Counter()
    for _, g in d.groupby(group):
        ents = sorted(zip(g.entity_text_normalised, g.entity_type))
        for (a, ta), (b, tb) in itertools.combinations(ents, 2):
            edges[(a, ta, b, tb)] += 1
    out = pd.DataFrame([(*k, w) for k, w in edges.items()],
                       columns=["source", "source_type", "target", "target_type", "weight"])
    return out[out.weight >= min_weight].sort_values("weight", ascending=False).reset_index(drop=True)

edges_news = cooccurrence(ner_news)
print(len(edges_news), "edges")
edges_news.head(20)
```

136149 edges

	source	source_type	target	target_type	weight
0	kentucky	geographic_location	lexington	geographic_location	982
1	fayette county	geographic_location	lexington	geographic_location	731
2	kentucky gazette	organization	lexington	geographic_location	711
3	frankfort	geographic_location	lexington	geographic_location	671
4	lexington	geographic_location	philadelphia	geographic_location	601
5	john bradford	person	lexington	geographic_location	578
6	danville	geographic_location	lexington	geographic_location	422
7	lexington	geographic_location	ohio	geographic_location	395
8	lexington	geographic_location	woodford county	geographic_location	360
9	lexington	geographic_location	virginia	geographic_location	329
10	bourbon county	geographic_location	lexington	geographic_location	309
11	levi todd	person	lexington	geographic_location	259
12	georgetown	geographic_location	lexington	geographic_location	251
13	lexington	geographic_location	washington	geographic_location	248
14	lexington	geographic_location	woodford	geographic_location	237
15	baltimore	geographic_location	philadelphia	geographic_location	234
16	kentucky gazette	organization	levi todd	person	229
17	fayette	geographic_location	lexington	geographic_location	226
18	france	geographic_location	paris	geographic_location	224
19	england	geographic_location	france	geographic_location	224

```
# The neighbourhood of one person: everyone named in the same paragraphs
def neighbours(name, edges, n=20):
    e = edges[(edges.source == name) | (edges.target == name)].copy()
    e["other"] = np.where(e.source == name, e.target, e.source)
    e["other_type"] = np.where(e.source == name, e.target_type, e.source_type)
    return e[["other", "other_type", "weight"]].head(n)

top_person = ner_news[ner_news.entity_type == "person"]["entity_text_normalised"].value_counts().index[0]
print(top_person); neighbours(top_person, edges_news)
```

```
john bradford
```

	other	other_type	weight
5	lexington	geographic_location	578
28	kentucky gazette	organization	193
122	john coburn	person	94
185	frankfort	geographic_location	80
221	alex. parker	person	72
268	james parker	person	67
322	kentucky	geographic_location	61
341	the kentucky gazette	organization	60
390	k e n t u c k y g a z e t t e.	organization	55
448	alexander	person	51
462	danville	geographic_location	50
794	philadelphia	geographic_location	35
817	alexander & james parker	organization	33
975	versailles	geographic_location	31

7.3 Typed relations by pattern

The pension clerk wrote the same sentences a hundred thousand times. A few patterns turn them into typed edges: `(file) —widow_of→ (person)`, `(file) —served_under→ (person)`, `(file) —resident_of→ (place)`. Newspapers get `reprinted_from` (Cordell's relation) and obituary formulae.

```
NAME = r"((?:(?:Capt|Col|Gen|Maj|Lieut|Dr|Mr|Mrs)\.?\s+)?[A-Z][a-z]+(?:\s+[A-Z]\.?)?(?:\s+[A-Z][a-z]+){1,2})"
PLACE = r"([A-Z][a-z]+(?:\s+[A-Z][a-z]+)?(?:\s+(?:county|County))?)"

RELATIONS = {
    # pension register
    "widow_of":      rf"\b(?:widow|relict)\s+of\s+(?:the\s+said\s+)?{NAME}",
    "child_of":      rf"\b(?:son|daughter)\s+of\s+{NAME}",
    "married":       rf"\b(?:was\s+)?married\s+(?:to\s+)?{NAME}",
    "served_under":  rf"\b(?:under|commanded\s+by)\s+(?:the\s+command\s+of\s+)?{NAME}",
    "resident_of":   rf"\b(?:resident|inhabitant)\s+of\s+(?:the\s+)?(?:county\s+of\s+)?{PLACE}",
    "late_of":       rf"\blate\s+of\s+{PLACE}",
    "died_at":       rf"\b(?:died|killed|wounded)\s+(?:at|in|near)\s+{PLACE}",
    # newspaper register
    "reprinted_from": r"\bFrom\s+the\s+([A-Z][A-Za-z\-]+(?:\s+[A-Z][A-Za-z\-]+)?\s+(?:Gazette|Advertiser|Journal|Herald|Chronicle|Star|Mercury|Centinel|Sent
    "consort_of":    rf"\b(?:consort|wife)\s+of\s+{NAME}",
}

def extract_relations(df, text_col="block_text"):
    """One row per (document, relation, object). Deduplicates repeated block_text first."""
    d = df.drop_duplicates("window_id")[["collection", "record_id", "page", "window_id", text_col]]
    rows = []
    for _, r in d.iterrows():
        txt = re.sub(r"\s+", " ", r[text_col] or "")
        for rel, rx in RELATIONS.items():
            for m in re.finditer(rx, txt):
                rows.append({"collection": r.collection, "record_id": r.record_id, "page": r.page,
                             "relation": rel, "object": m.group(1).strip(),
                             "evidence": txt[max(0, m.start()-60): m.end()+60]})
    return pd.DataFrame(rows)

rel_pens = extract_relations(ner_pens)
rel_news = extract_relations(ner_news)
print(len(rel_pens), "pension relations;", len(rel_news), "newspaper relations")
pd.concat([rel_pens, rel_news])["relation"].value_counts()
```

```
4825 pension relations; 175 newspaper relations
```

relation	count
served_under	2403
widow_of	910
resident_of	721
married	247
late_of	239
died_at	226
child_of	135
consort_of	100
reprinted_from	19

dtype: int64

```
# Always read the evidence column before trusting a pattern
rel_pens[rel_pens.relation == "widow_of"].head(10)[["record_id", "object", "evidence"]]
```

	record_id	object	evidence
8	111404146	Ansel Converse	gton in the County and State aforesaid, Olive Converse, the widow of Ansel Converse of Butternuts in the State of New York, and Nancy Shaw, the
11	111404093	Jonathan Torrey	ting half pay & pensions to certain widows" that she is the widow of Jonathan Torrey who was a private in the army of the Revolution, that the s
13	111403929	Amos Torry Conn	widow of Amos Torry Conn.
41	111404196	Nathl B Toney Pens	Sarah Toney widow of Nathl B Toney Pens^a - act 38.43.244 New York act 48 - 3747
48	111403929	Daniel Culver	In the case of ~~A~~ the heirs of Abigail Culver, widow of Daniel Culver, presented by me, no proof of identity was required or give
50	111404196	Nathaniel B. Torrey	ns who served during the Revolutionary war, that she is the widow of Nathaniel B. Torrey, who was a private in the revolution, and that she received
63	111404292	Samuel Torrey	half pay and pensions to certain Widows" - that she is the widow of Samuel Torrey, formerly of said Sutton, who was a private soldier in the
67	111404196	Nathaniel B. Maj	3747 City New York ~~Albany~~ Sarah Torrey, widow of Nathaniel B. Maj who served in the Revolutionary war, as a private
68	111404292	Samuel Mass	Hannah Sorey widow of Samuel Mass.
69	111403929	Amos Torry	Sarah Torry widow of Amos Torry. the said Torry a Revolutionary soldier. Conn. line ~~pensi

```
# Which newspapers reprinted whom? (Cordell's relation)
rel_news[rel_news.relation == "reprinted_from"]["object"].value_counts().head(15)
```

object	count
New-York Daily Advertiser	2
London Gazette	2